

$ab=ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$
 $\frac{\mathbb{Z} \times \mathbb{Z}}{\mathbb{Z} \times \mathbb{Z}}$
 $S^1 \vee S^2$
 $T = \mathbb{Z} \times \mathbb{Z}$
 $T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$
 $K \hookrightarrow \mathbb{Z} \times \mathbb{Z}$

[illegible]

$\mathbb{R}/\mathbb{P}$: 0-cell: $\{v\}$
 1-cell: $\{x, y\}$
 2-cell: $\{z\}$

[illegible]

Hand-drawn sketches of various mathematical concepts:

- A sphere with a point on its surface.
- A circle with a point on its circumference.
- A series of circles arranged in a row.
- A flower-like shape with five petals.
- A spiral shape.
- A horizontal line with several points marked on it.
- A 3D shape, possibly a cube or a rectangular prism, with arrows indicating dimensions.
- A U-shaped curve.